Diket: $a = 3$ dan $b = 2$

A. Matriks A Eksplisit dan Determinan

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} \xrightarrow{\text{subsitusi}} \begin{bmatrix} 3+1 & 2 & 2 \\ 1 & 3+2 & 2 \\ 2 & 1 & 3+3 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 1 & 5 & 2 \\ 2 & 1 & 6 \end{bmatrix}$$

Hitung det (A) dgn ekspansi kofaktor baris pertama:

$\det(A) = a_{11}(C_{11}) - a_{12}(C_{12}) + a_{13}(C_{13})$

$$\det(A) = 4 \begin{vmatrix} 5 & 2 \\ 1 & 6 \end{vmatrix} - 2 \begin{vmatrix} 1 & 2 \\ 2 & 6 \end{vmatrix} + 2 \begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix}$$

$\det(A) = 4(30-2) - 2(6-4) + 2(1-10)$

$\det(A) = 4(28) - 2(2) + 2(-9)$

$\det(A) = 112 - 4 - 18 = 90$

B. Matriks Minor ($M_{ij}$)

Minor $M_{ij}$ adalah determinan submatriks setelah menghapus baris i dan kolom j

| i/j | 1 | 2 | 3 |
|---|---|---|---|
| 1 | $M_{11} = (30-2) = 28$ | $M_{12} = (6-4) = 2$ | $M_{13} = (1-10) = -9$ |
| 2 | $M_{21} = (12-2) = 10$ | $M_{22} = (24-4) = 20$ | $M_{23} = (4-4) = 0$ |
| 3 | $M_{31} = (4-10) = -6$ | $M_{32} = (8-2) = 6$ | $M_{33} = (20-2) = 18$ |

C. Matriks Kofaktor (C):

$$C = \begin{bmatrix} 28 & -2 & -9 \\ -10 & 20 & 0 \\ -6 & -6 & 18 \end{bmatrix}$$

Matriks Adjoin ($Adj(A) = C^T$):

$$Adj(A) = \begin{bmatrix} 28 & -10 & -6 \\ -2 & 20 & -6 \\ -9 & 0 & 18 \end{bmatrix}$$

Matriks Invers ($A^{-1}$):

$$A^{-1} = \frac{1}{90} \begin{bmatrix} 28 & -10 & -6 \\ -2 & 20 & -6 \\ -9 & 0 & 18 \end{bmatrix}$$

E. Studi Kasus Citra

Blok piksel asli P (N = 32):

Subsitusi $N = 32$ dan $a = 3$ ke dlm matriks P:

- $50 + N = 50 + 32 = 82$
- $a + 2 = 3 + 2 = 5$
- $10 + N = 10 + 32 = 42$
- $30 + N = 30 + 32 = 62$

$$P = \begin{bmatrix} 120 & 82 & 200 \\ 80 & 5 & 42 \\ 62 & 90 & 170 \end{bmatrix}$$

Proses enskripsi ($E = A \cdot P$)

Elemen $E_{11}$: $(4 \cdot 120) + (2 \cdot 80) + (2 \cdot 62) = 480 + 160 + 124 = 764$

Dipindai dengan CamScanner

Setelah menghitung seluruh elemen, akan mendapatkan matriks terenkripsi E yg nilainya jauh di atas rentang grayscale (0-255), itulah sebabnya dlm visualisasi Matlab digunnakan fungsi imshow(E, []) untk melakukan scaling otomatis.

Dipindai dengan CamScanner